

AVR Based Programmable Object Counter



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Object counters are used for applications like counting the number of visitors at super markets and other public places, the number of objects passing through a

particular area or into a tray or container in manufacturing industries, the number of tablets filled in a bottle or a wrapper in the pharmaceutical industry and counting and limiting the number of people entering elevators or halls, among others.

The project presented here demonstrates the working of an object counter suitable for industrial applications for counting up to 9999 objects. It is an AVR microcontroller (MCU) based full-featured programmable counter. Its prototype on breadboard is shown in Fig. 1.

In manufacturing industries, before the final product is ready, the product passes through a number of different stages of the manufacturing process. Normally, the operator enters the number of objects to be counted from the monitoring con-

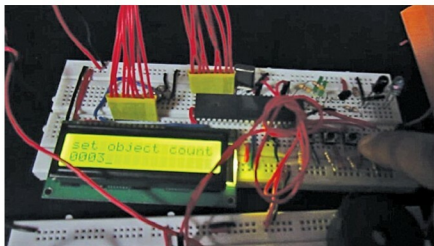


Fig. 1: Prototype on breadboard

Test Points

Test point	Details
TP0	0V (GND)
TP1	5V
TP2	12V when S6 is closed
TP3	Low when object is detected

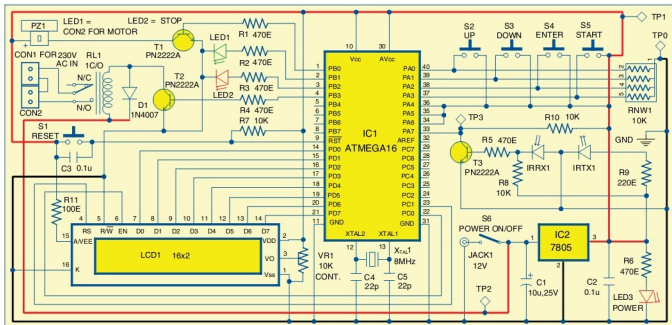


Fig. 2: Circuit diagram of the AVR based programmable object counter